UNIVERSITI TEKNIKAL MALAYSIA MELAKA

INDIVIDUAL LAB REPORT

PROGRAM

BITA GROUP 2 SEMESTER 1

ASSESSMENT

LAB 5QS

SUBJECT

BITP1113

LECTURER

DR NUR AUNI

DATE OF SUBMISSION

[29 / 10 / 2026]

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EXERCISE 4.3

Question 1

```
int main() {
int number;

cout << "Enter a number: ";
cin >> number;

if (number > 0) {
    cout << "The number you entered is " << number
        << ", and this is a positive number." << endl;
}
else if (number < 0) {
    cout << "The number you entered is " << number
        << ", and this is a negative number." << endl;
}
else {
    cout << "The number you entered is " << number
        << ", and this is a zero." << endl;
}

return 0;
}
```

```
Enter a number: 4
The number you entered is 4, and this is a positive number.

C:\Users\amiru\source\repos\LAB5\x64\Debug\LAB5.exe (process 11836) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```

Question 2

```cpp
#include <iostream>
using namespace std;

int main() {
    double cost, rate, estimate_cost;
    int years;

    cout << "Enter the current cost: RM";
    cin >> cost;

    cout << "Enter inflation rate in percentage (e.g. 2.5%)
    cin >> rate;

    cout << "Enter the number of years: ";
    cin >> years;

    rate = rate / 100;
    estimate_cost = cost;

    for (int i = 1; i <= years; i++) {
        cout << "i = " << i << endl;

        cout << "estimate cost before = "
            << estimate_cost << endl;

        estimate_cost = estimate_cost * (1 + rate);

        cout << "estimate cost after = "
            << estimate_cost << endl;
    }

    cout << "Estimate cost after " << years
        << " year(s): RM" << estimate_cost << endl;

    return 0;
}
```

```
Enter the current cost: RM10
Enter inflation rate in percentage (e.g. 2.5%): 1.4%
Enter the number of years: Estimate cost after 0 year(s): RM10
```

Question 3

```cpp
#include <iostream>

using namespace std;

int main() {
    cout << "Prime numbers between 3 and 100 are:\n";

    // Outer loop iterates from 3 to 100
    for (int n = 3; n <= 100; n++) {
        bool isPrime = true;

        // Inner loop checks if n is prime by checking divisors from 2 to n-1
        for (int i = 2; i < n; i++) {
            if (n % i == 0) {
                isPrime = false; // n is evenly divisible, so it's not prime
                break;           // Exit the inner loop early
            }
        }

        // If no values from 2 to n-1 divided n, it is prime
        if (isPrime) {
            cout << n << " ";
        }
    }

    cout << endl;
    return 0;
}
```

```
Prime numbers between 3 and 100 are:
3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97
```